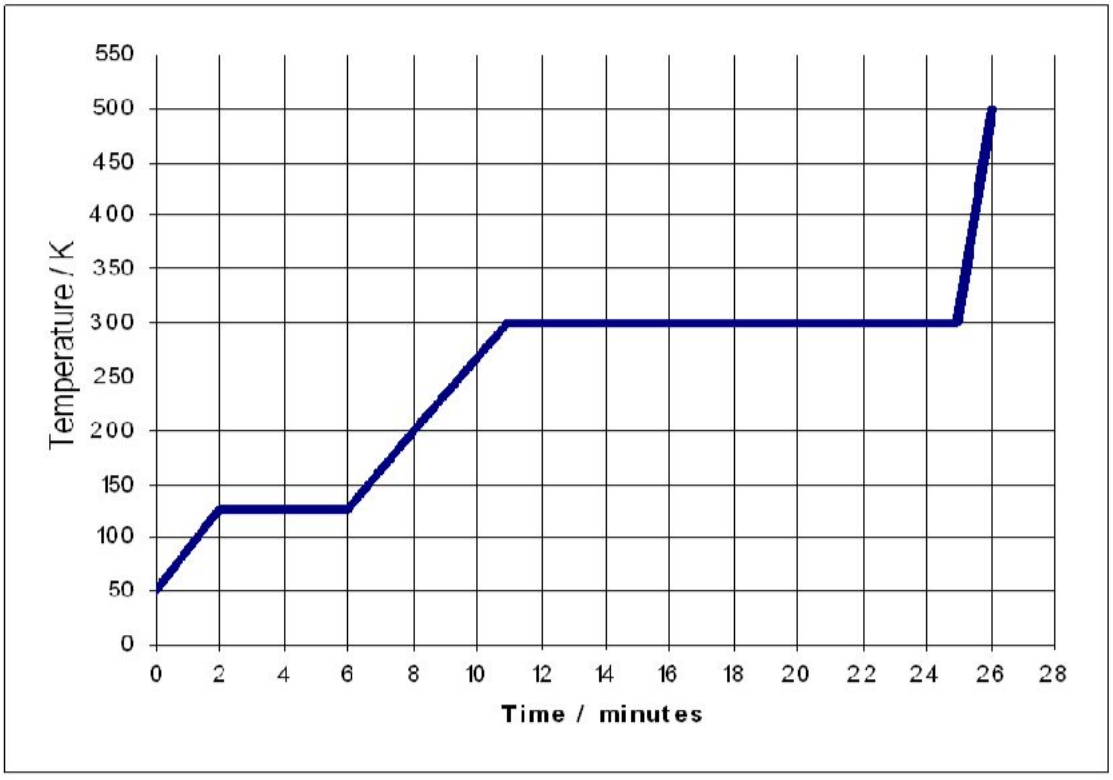
15 Heat is added to a pure substance in a closed container at a constant rate. A graph of the temperature of the substance as a function of time is shown below.



If L_f is the latent heat of fusion and L_v is the latent heat of vaporization of the substance, what is the value of the ratio $\frac{L_v}{L_f}$ for this substance?

A 1.0 **B** 2.4 **C** 3.5 **D** 5.0

Ans: C

Heat energy required for melting, $Q_1 = mL_f$

Heat energy required for vaporization, $Q_2 = mL_v$

Let P represent the constant rate of heat supply.

Required ratio, $\frac{L_v}{L_f} = \frac{Q_2/m}{Q_1/m} = \frac{Q_2}{Q_1} = \frac{Pt_2}{Pt_1} = \frac{t_2}{t_1} = \frac{25-11}{6-2} = 3.5$